

# Generation Interconnection System Impact Study Report

## For Queue Project Q18-1056

Mt Gate Junction 100 kV

December 28, 2019

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1056. The Interconnection Customer proposes a 179.1 MW solar photovoltaic facility interconnecting at the Mt Gate Junction 100 kV substation in the Illinois area. The request was received on September 02, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 384 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,924,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                 | Contingency                      | Loading (%) | Rating (MVA) | DFAX (%) |
|----------------------------------------------------|----------------------------------|-------------|--------------|----------|
| MT GATE JUNCTION / ROLLINS 100kV line              | FORK PICKETT / PLANO 100kV       | 113.7       | 137          | 19.4     |
| EASTOWN RD / SWAN VLY 230kV L                      | SWANVLY / TREGO 500kV            | 110.6       | 1,027        | 6.3      |
| POST ROCK / NEWPORT TIE 230kV                      | TREGO / BEAVER 500kV             | 110.0       | 2,343        | 13.2     |
| ADIRONDACK 500-230 kV TX                           | E METHUEN 500/230 kV Transformer | 109.7       | 1,620        | 5.1      |
| SLOCUM RD SUBSTATION / MCNEIL 230kV                | RICHLAND / DEKALB 500kV          | 109.1       | 2,032        | 25.6     |
| POST ROCK / MYRTLE POINT 230kV                     | TREGO / BEAVER 500kV             | 108.9       | 2,391        | 12.8     |
| SLOCUM RD / DEKALB 230kV line                      | RICHLAND / DEKALB 500kV          | 108.8       | 2,066        | 25.6     |
| LARSON ST / NORWICH 230kV                          | NORWICH / BARRETT 500kV          | 108.5       | 1,633        | 8.9      |
| MINNEHAHA / SIDNEY A MURRAY JR HYDROELECTRIC 230kV | RICHLAND / DEKALB 500kV Line     | 108.4       | 1,403        | 16.4     |
| MCNEIL / BREED 230kV Line                          | RICHLAND / DEKALB 500kV          | 108.4       | 2,060        | 25.6     |
| BREED / SIDNEY A MURRAY JR HYDROELECTRIC 230kV     | DEKALB / RICHLAND 500kV          | 108.4       | 2,060        | 25.6     |
| NEW DEPARTURE / ADAMS 230kV                        | TREGO / BEAVER 500kV line        | 107.4       | 1,433        | 7.8      |
| MASON ST 500 kV/230 kV Transformer Bank            | BEAVER / TREGO 500kV L           | 107.3       | 2,151        | 11.9     |
| HERALD-TWIN CITIES / MASON ST SUBSTATION 230kV     | MASON ST / NEW DEPARTURE 230kV L | 107.3       | 936          | 5.5      |
| INOLTA / OLD FORGE 230kV                           | HYLEBOS / RAVENNA 500kV          | 107.2       | 520          | 5.3      |
| IMPERIAL / GRUVER 230kV Line                       | IMPERIAL / EUFAULA 500kV         | 107.0       | 779          | 6.6      |

| Monitored Facility                                 | Contingency                              | Loading (%) | Rating (MVA) | DFAX (%) |
|----------------------------------------------------|------------------------------------------|-------------|--------------|----------|
| BEAVER / ONANCOCK DIESEL GENERATION FACILITY 230kV | BEAVER / EUFAULA 500kV L                 | 106.3       | 2,230        | 10.8     |
| EUFAULA 230/1kV Transformer Bank                   | IMPERIAL SUBSTATION / EUFAULA 500kV      | 105.8       | 2,552        | 12.3     |
| EUFAULA 500/1kV TX                                 | IMPERIAL SUBSTATION / EUFAULA 500kV line | 105.8       | 2,552        | 12.3     |
| HIGHGATE / HERALD-TWIN CITIES 230kV                | MASON ST 500 kV/230 kV Transformer       | 105.1       | 1,416        | 7.0      |
| JOSO / EUFAULA 230kV L                             | BEAVER / EUFAULA 500kV line              | 104.9       | 1,073        | 8.1      |
| 24TH & GARDENDALE SUBSTATION / LANDSMAN CRK 230kV  | RICHLAND / DEKALB 500kV                  | 104.8       | 832          | 9.1      |
| BEAVER 230/1 kV Autotransformer                    | EUFAULA / BEAVER 500kV                   | 104.8       | 3,059        | 13.7     |
| BEAVER 500 kV/1 kV Transformer                     | BEAVER / EUFAULA 500kV                   | 104.8       | 3,059        | 13.7     |
| MUSKOGEE SUBSTATION / SWAN VLY 500kV line          | RICHLAND / DEKALB 500kV                  | 104.7       | 4,382        | 23.4     |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                           | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| MT GATE JCT / ROLLINS 100kV                        | Rebuild line with higher-capacity conductor | \$8,404,000   | 100% (\$8,404,000)    |
| EASTOWN RD / SWAN VALLEY 230kV                     | Rebuild line with higher-capacity conductor | \$9,539,000   | 1% (\$124,000)        |
| POST ROCK / NEWPORT TIE 230kV                      | Reconductor line                            | \$10,955,000  | 1% (\$142,000)        |
| ADIRONDACK 500-230 kV TX                           | Replace transformer with higher-rated unit  | \$30,612,000  | 1% (\$398,000)        |
| SLOCUM RD SUBSTATION / MCNEIL 230kV                | Reconductor line                            | \$4,060,000   | 1% (\$53,000)         |
| POST ROCK / MYRTLE POINT 230kV                     | Reconductor line                            | \$9,926,000   | 1% (\$129,000)        |
| SLOCUM RD / DEKALB 230kV line                      | Reconductor line                            | \$15,069,000  | 1% (\$195,000)        |
| NORWICH / LARSON TREET 230kV Line                  | Reconductor line                            | \$3,910,000   | 1% (\$51,000)         |
| MINNEHAHA / SIDNEY A MURRAY JR HYDROELECTRIC 230kV | Reconductor line                            | \$12,889,000  | 1% (\$167,000)        |
| MCNEIL / BREED 230kV Line                          | Reconductor line                            | \$6,929,000   | 1% (\$90,000)         |
| BREED / SIDNEY A MURRAY JR HYDROELECTRIC 230kV     | Reconductor line                            | \$13,336,000  | 1% (\$173,000)        |
| NEW DEPARTURE / ADAMS 230kV                        | Reconductor line                            | \$5,307,000   | 1% (\$69,000)         |

| Facility                                           | Required Upgrade                           | Cost Estimate | Allocation to Project |
|----------------------------------------------------|--------------------------------------------|---------------|-----------------------|
| MASON ST 500 kV/230 kV Auto                        | Replace transformer with higher-rated unit | \$24,654,000  | 1% (\$320,000)        |
| MASON ST / HERALD-WIN CITIES 230kV                 | Reconductor line                           | \$17,479,000  | 1% (\$227,000)        |
| OLD FORGE / INOLTA 230kV                           | Reconductor line                           | \$8,444,000   | 1% (\$110,000)        |
| IMPERIAL / GRUVER 230kV Line                       | Reconductor line                           | \$3,135,000   | 1% (\$41,000)         |
| ONANCOCK DIESEL GENERATION FACILITY / BEAVER 230kV | Reconductor line                           | \$7,759,000   | 1% (\$101,000)        |
| EUFAULA 230/1kV Transformer Bank                   | Replace transformer with higher-rated unit | \$6,504,000   | 1% (\$84,000)         |
| EUFAULA 500/1kV TX                                 | Replace transformer with higher-rated unit | \$29,302,000  | 1% (\$380,000)        |
| HERALD-TWIN CITIES / HIGGATE 230kV                 | Reconductor line                           | \$8,393,000   | 1% (\$109,000)        |
| EUFULA / JOSO 230kV                                | Reconductor line                           | \$2,999,000   | 1% (\$39,000)         |
| 24TH & GARDENDALE / LANDSMAN CRK 230kV             | Reconductor line                           | \$13,904,000  | 1% (\$180,000)        |
| BEAVER 230 kV/1 kV Autotransformer                 | Replace transformer with higher-rated unit | \$11,336,000  | 1% (\$147,000)        |
| BEAVER 500 kV/1 kV Transformer                     | Replace transformer with higher-rated unit | \$27,535,000  | 1% (\$357,000)        |
| SWAN VLY / MUSOGEE 500kV                           | Reconductor line                           | \$64,346,000  | 1% (\$834,000)        |
|                                                    | Total allocated to project                 |               | \$12,924,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.